

Data-Efficient Preference-Aligned Robotic Shelf Arrangement with Composable Constraint Models

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We study the problem of data-efficient object arrangement on shelves with existing objects, where a robot must propose geometrically feasible and semantically aligned goal configurations for newly introduced items while optionally rearranging existing ones. The task is inherently hierarchical: decisions must be made at the level of shelf layers and compartments before resolving precise local object poses. We propose a unified framework that combines (i) a hierarchical generative model that samples geometrically and physically valid shelf configurations, (ii) composable diffusion or energy-based guidance to enforce constraints and minimize rearrangement effort, and (iii) a semantic alignment module driven by large language models (LLMs) and preference diffusion to capture user habits and commonsense organization principles. Semantic alignment is formalized through a semantic projection and distance defined over configurations, enabling reward-like gradient guidance without requiring dense scalar rewards. We further introduce a continual preference learning mechanism that adapts to evolving user feedback while retaining stability via replay and regularization. Our formulation explicitly accounts for compartment capacity, rearrangement feasibility, and minimal-disruption objective when assigning new objects to shelf compartments. We outline a data generation pipeline and training strategy that leverage synthetic sampling, weak supervision from LLMs, and limited human feedback, achieving strong generalization with minimal data. The proposed approach unifies hierarchical planning, generative modeling, and preference-based learning, offering a scalable and flexible solution for personalized robotic shelf organization.

Additional Key Words and Phrases: robot manipulation, scene rearrangement, preference learning, diffusion models, energy-based models, language models

1 Introduction

Robotic object placement in human environments is a long-standing challenge at the intersection of robotics, computer vision, and machine learning. Among these problems, shelf organization is particularly difficult due to its inherently *hierarchical*, *semantic*, and *combinatorial* nature. A shelf typically consists of multiple layers and finer-grained compartments, each implicitly associated with user-specific habits, commonsense expectations, and spatial affordances. When new objects must be added to an already populated shelf, a robot must not only identify feasible geometric placements, but also reason about rearranging existing objects to create space while preserving semantic coherence.

Prior work has largely focused on either geometric feasibility or end-to-end learning-based placement policies. However, real-world shelf organization requires simultaneously satisfying: (i) *geometric and physical constraints*, such as non-overlap, support stability, and shelf capacity; (ii) *semantic constraints*, such as grouping similar objects, respecting usage frequency, or maintaining intuitive global layouts; and (iii) *user preferences*, which may vary across individuals and evolve over time. Moreover, naively greedily placing objects often leads to suboptimal or semantically incoherent configurations, especially when rearrangement of existing objects is allowed.

In this work, we focus on the second variant of the shelf arrangement problem: given a shelf with existing objects, generate a goal configuration that places additional objects while optionally rearranging existing ones. We explicitly model shelf organization as a structured configuration space, and seek a goal state that lies within the geometrically valid subspace while minimizing semantic deviation and maximizing preference compliance.

To achieve this, we propose a **hierarchical, composable generative framework** that decouples geometry from semantics. A diffusion-based or energy-based generative model is used to sample geometrically and physically valid

configurations at multiple levels of abstraction, from compartment assignment to precise object poses. Semantic alignment is enforced through a learned semantic projection and preference diffusion model, which provides reward-like gradient guidance without collapsing preferences into a single scalar reward. Large language models (LLMs) are incorporated to inject commonsense priors, guide compartment-level decisions, and improve data efficiency during preference model training.

Our approach offers several advantages: it supports data-efficient learning, allows flexible composition of constraints, enables continual adaptation to user feedback, and naturally balances minimal rearrangement effort with semantic preservation. By explicitly separating *where* an object should go (semantic and hierarchical reasoning) from *how* it can be placed (geometric feasibility), the proposed method provides a scalable and interpretable solution to real-world shelf organization.

2 Related Work

2.1 Robotic Object Placement and Rearrangement

Early work on robotic placement emphasized geometric reasoning, stability analysis, and collision avoidance. Sampling-based planners and optimization methods have been widely used to ensure physical feasibility in cluttered scenes [5, 9]. Rearrangement planning extends this setting by allowing objects to be temporarily displaced to enable goal placements [15, 22]. While effective for small-scale problems, these methods typically lack semantic awareness and do not scale well to complex, preference-driven environments.

Learning-based approaches have introduced data-driven priors for object placement. Supervised models and reinforcement learning have been applied to predict placement locations based on visual input [7, 27]. However, these approaches often require large datasets and struggle to generalize across users, shelf layouts, and object categories.

2.2 Semantic and Commonsense Reasoning for Placement

Semantic reasoning has been explored through object affordances, functional relations, and contextual cues [14, 23]. More recent works incorporate graph representations or relational models to encode object-object and object-environment relationships [18, 24]. Despite these advances, semantic constraints are often treated as hard-coded rules or static scores, limiting adaptability to individual user habits.

Large language models have recently emerged as powerful tools for encoding commonsense knowledge and spatial reasoning [1, 11]. LLMs have been used to guide planning, generate high-level task decompositions, and evaluate candidate solutions. However, directly relying on LLM outputs for low-level control or geometric reasoning remains challenging, motivating hybrid systems where LLMs provide semantic guidance while learned models handle physical feasibility.

2.3 Diffusion Models and Energy-Based Models for Structured Generation

Diffusion models have demonstrated strong performance in generating complex, multimodal distributions [10, 21]. Recent work has extended diffusion models to structured outputs such as trajectories, scenes, and graphs [12, 28]. Energy-based models (EBMs) offer an alternative formulation, enabling the composition of multiple constraints through additive energy terms [6, 17]. Both paradigms are well-suited for constraint satisfaction problems where exact likelihoods are intractable.

Composable diffusion and EBMs are particularly attractive for shelf organization, as they allow geometric, physical, and semantic constraints to be combined without retraining a monolithic model [8, 26]. Our work builds on this insight by using a generative model to define the space of valid configurations, while semantic alignment is imposed via gradient-based guidance.

2.4 Preference Learning and Human-in-the-Loop Alignment

Learning from human preferences has gained significant attention as an alternative to explicit reward engineering. Pairwise comparison models and preference-based reinforcement learning have been successfully applied in robotics and control [3, 20]. More recently, preference learning has been combined with diffusion-style score matching to provide reward-like gradients without collapsing preferences into a single scalar objective [25].

Continual and lifelong learning of preferences is critical in personalized environments. Replay buffers, regularization, and stochastic dynamics have been used to mitigate catastrophic forgetting [2, 16]. In this work, we adopt a preference diffusion formulation with an online parameter evolution ODE, enabling stable adaptation to evolving user habits.

3 Problem Statement

We consider the problem of *goal configuration generation* for robotic shelf organization under geometric, physical, and semantic constraints. Our focus is on the second variant of shelf organization tasks, which is both practically relevant and algorithmically challenging: placing new objects onto a shelf that already contains existing objects, while optionally rearranging the existing objects to create space and preserve semantic coherence.

3.1 Task Variants and Scope

Object placement on shelves can be broadly categorized into three variants:

- (1) Placing a set of objects on an empty shelf.
- (2) Placing one or more new objects onto a shelf that already contains objects.
- (3) Rearranging all objects on a shelf to improve organization or aesthetics.

While the first variant primarily concerns packing and geometric feasibility, and the third variant emphasizes global reorganization, the second variant requires balancing *incremental insertion*, *minimal disruption*, and *semantic preservation*. This setting arises frequently in domestic and retail environments, where shelves evolve over time rather than being reorganized from scratch [13, 15].

In this work, we explicitly address the second variant.

3.2 Configuration Space Representation

Let the set of objects initially present on the shelf be

$$O_{\text{init}} = \{o_1, o_2, \dots, o_n\},$$

with corresponding positions

$$P_{\text{init}} = \{p_1, p_2, \dots, p_n\}.$$

The initial shelf configuration is represented as

$$s_{\text{init}} = ((o_1, p_1), (o_2, p_2), \dots, (o_n, p_n)), \quad (1)$$

where each position p_i denotes the pose of object o_i in a shelf-centric coordinate system. We intentionally keep the representation abstract; object attributes (e.g., geometry, mass, category) and pose parameterizations can be instantiated differently depending on the sensing and control stack.

Given a set of new objects

$$O_{\text{new}} = \{o_{n+1}, o_{n+2}, \dots, o_{n+m}\},$$

we define an augmented initial configuration

$$\begin{aligned} s'_{\text{init}} &= s_{\text{init}} \oplus s_{\text{new}} \\ &= ((o_1, p_1), \dots, (o_n, p_n), (o_{n+1}, \text{null}), \dots, (o_{n+m}, \text{null})), \end{aligned} \quad (2)$$

where null indicates that the object has not yet been assigned a shelf pose.

3.3 Goal Configuration and Rearrangement

The objective is to find a goal configuration

$$\begin{aligned} s_{\text{goal}} &= ((o_1, p'_1), \dots, (o_n, p'_n), (o_{n+1}, p'_{n+1}), \dots, (o_{n+m}, p'_{n+m})) \\ &= F_{\text{move}}(s_{\text{init}}) \oplus s_{\text{add}}, \end{aligned} \quad (3)$$

where F_{move} denotes a (possibly empty) rearrangement of existing objects, and s_{add} denotes the placements of newly added objects. Unlike approaches that assume fixed initial placements, we explicitly allow F_{move} to relocate existing objects, reflecting the fact that feasible and semantically meaningful insertion often requires limited rearrangement [4, 22].

3.4 Geometric and Semantic Constraints

Let S denote the configuration space of all possible object–pose assignments, and let $S_{\text{valid}} \subset S$ denote the subset of configurations that satisfy geometric and physical constraints, including:

- non-overlapping object placements,
- containment within shelf geometry,
- static stability and support constraints.

These constraints are necessary but not sufficient for human-acceptable shelf organization.

Beyond geometric validity, we define *semantic constraints* that capture how humans perceive and evaluate shelf organization. These include both local relations (e.g., grouping similar objects, maintaining visual order within a compartment) and global relations (e.g., placing frequently used items at accessible heights, separating incompatible object categories). Such semantic considerations have been shown to play a critical role in human judgments of object arrangement [13, 19].

3.5 Semantic Projection and Optimization Objective

To formalize semantic alignment, we define:

- a semantic projection function

$$f_{\theta_1} : S \rightarrow A,$$

mapping configurations to an abstract semantic space A ;

- a semantic distance function

$$D_{\theta_2}(s_1, s_2) = D_{\theta_2}(f_{\theta_1}(s_1), f_{\theta_1}(s_2)),$$

measuring how different two configurations appear semantically;

- a compliance function

$$g_{\theta_3} : A \rightarrow \mathbb{R},$$

scoring alignment with commonsense and user-specific habits.

Given the augmented initial configuration s'_{init} , our goal is to find a configuration $s_{\text{goal}} \in S_{\text{valid}}$ that jointly: (i) minimizes semantic deviation from the original arrangement, (ii) maximizes semantic compliance with user preferences, and (iii) avoids unnecessary rearrangement. Formally, we seek

$$s^* = \arg \min_{s \in S_{\text{valid}}} \left[D_{\theta_2}(s, s_{\text{init}}) - \lambda g_{\theta_3}(f_{\theta_1}(s)) + \beta C_{\text{move}}(s, s_{\text{init}}) \right], \quad (4)$$

where C_{move} penalizes the magnitude of rearrangement and λ, β balance semantic alignment and minimal disruption.

This formulation captures the core challenge of incremental shelf organization: generating physically feasible goal states that preserve semantic structure and adapt to evolving user preferences, without resorting to complete reorganization or brittle hand-crafted rules.

4 Proposed Method

We propose a hierarchical, data-efficient framework for generating object placement and rearrangement goals on a multi-compartment shelf. The method combines (i) a geometry- and physics-aware generative model that produces valid configurations, (ii) a semantic alignment module guided by preference diffusion, and (iii) a large language model (LLM) that injects commonsense priors and improves data efficiency. Our approach explicitly accounts for hierarchical decision structure, incremental object insertion, and minimal-effort rearrangement of existing objects.

4.1 Problem Structure and Hierarchy

Given an initial shelf configuration s_{init} (Eq. 1) and a set of new objects O_{new} , the task exhibits a natural hierarchy:

- (1) **Global decision:** which shelf layer and smaller compartment each new object should be assigned to;
- (2) **Local decision:** precise continuous pose (position and orientation) of each object within the chosen compartment;
- (3) **Rearrangement decision:** how existing objects should be minimally displaced to create sufficient space.

We exploit this hierarchy to factorize the search over configurations and improve both sample efficiency and controllability.

4.2 Geometric-Physical Generative Model

We assume access to a generative model $p_{\theta}(s)$ that defines a distribution over shelf configurations $s \in S_{\text{valid}}$, where S_{valid} denotes the subspace of configurations satisfying geometric and physical constraints (e.g., non-overlap, stability, shelf capacity).

The model is hierarchical:

$$p_{\theta}(s) = p_{\theta_1}(c) p_{\theta_2}(r \mid c) p_{\theta_3}(p \mid r, c), \quad (5)$$

where:

- c denotes discrete compartment assignments,
- r denotes rearrangement actions on existing objects,

- p denotes continuous object poses.

This factorization allows us to first sample feasible compartment-level allocations, then sample minimal rearrangements, and finally refine local placements.

4.3 Semantic Representation and Alignment

We model semantic structure through a learned projection

$$f_{\theta_1} : S \rightarrow A, \quad (6)$$

which maps configurations to a latent semantic space A capturing human-perceived organization (e.g., category grouping, accessibility, visual balance).

Semantic consistency between configurations is measured via a semantic distance:

$$D_{\theta_2}(s_1, s_2) = \|f_{\theta_1}(s_1) - f_{\theta_1}(s_2)\|, \quad (7)$$

and a compliance score:

$$g_{\theta_3}(s) = g_{\theta_3}(f_{\theta_1}(s)), \quad (8)$$

which evaluates alignment with user habits or commonsense organization.

4.4 Preference Diffusion for Semantic Guidance

Instead of collapsing semantic alignment into a single scalar reward, we adopt *preference diffusion* to model user judgments over object placement trajectories.

4.4.1 Preference Modeling. Let τ denote a trajectory from s_{init} to a candidate goal configuration s_{goal} . We learn:

$$p_{\phi}(\text{pref} \mid \tau), \quad (9)$$

or pairwise preferences:

$$p_{\phi}(\tau_A \succ \tau_B), \quad (10)$$

where preferences may encode accessibility, tidiness, or user-specific habits.

The resulting score function:

$$s_{\phi}(\tau) = \nabla_{\tau} \log p_{\phi}(\text{pref} \mid \tau), \quad (11)$$

acts as a semantic guidance field over configuration trajectories, analogous to reward gradients but without requiring explicit reward specification.

4.4.2 Continual Preference Adaptation. User feedback arrives incrementally. We update ϕ via a preference ODE:

$$\frac{d\phi}{dt} = -\eta \nabla_{\phi} L_{\text{pref}} - \lambda \nabla_{\phi} L_{\text{replay}} - \gamma(\phi - \phi_0) + \sigma \xi(t), \quad (12)$$

enabling lifelong adaptation while preserving stability.

4.5 LLM-Enhanced Semantic Efficiency

We incorporate an LLM in two complementary ways:

- (1) **Semantic scoring:** the LLM directly evaluates high-level compartment assignments (e.g., “books should be grouped by topic and placed at eye level”) and provides weak preference labels;

- (2) **Data augmentation:** the LLM generates synthetic pairwise preferences and rationales, significantly reducing the amount of human-labeled data needed to train p_ϕ .

This hybrid design combines neural perceptual grounding with symbolic commonsense priors.

4.6 Greedy Incremental Insertion with Minimal Rearrangement

To efficiently handle multiple new objects, we adopt a greedy but semantics-aware insertion strategy. At each step, we solve:

$$\min_{s \in S_{valid}} D_{\theta_2}(s, s_{init}) - \alpha g_{\theta_3}(s) + \beta C_{move}(s, s_{init}), \quad (13)$$

where C_{move} penalizes total rearrangement magnitude of existing objects.

Compartment selection explicitly accounts for:

- available free volume,
- estimated rearrangement feasibility,
- semantic compatibility with existing objects.

This ensures that greedily adding objects yields globally coherent configurations while avoiding unnecessary disruption.

4.7 Overall Optimization Objective

The final goal configuration is obtained by sampling from the geometry-aware generator and reweighting samples using preference diffusion:

$$p(s_{goal} | s_{init}) \propto p_\theta(s_{goal}) \exp(-D_{\theta_2}(s_{goal}, s_{init}) + g_{\theta_3}(s_{goal})). \quad (14)$$

This formulation unifies physical feasibility, semantic alignment, and user-specific preferences within a single probabilistic framework.

5 Data Generation and Training with Greedy Insertion

This section describes the unified data generation and training procedure for our framework. A key design principle is that the *training process mirrors the greedy insertion procedure used at inference time*. By generating and learning from incremental insertion trajectories, we ensure semantic preservation, minimal rearrangement, and geometric feasibility are enforced consistently across training and deployment.

5.1 Hierarchical Training and Inference with Constraint-Specific Generative Models

We adopt a fully compositional view of geometric and physical feasibility by modeling *each constraint with a separate generative or energy-based module*, and coupling these modules only at inference time. This design enables modularity, data efficiency, and flexible deployment under varying constraint sets.

5.1.1 Hierarchical Factorization. At each greedy insertion step t , the conditional distribution over successor configurations is factorized hierarchically:

$$p(s^{(t+1)} | s^{(t)}, o^{(t)}) = p_{\theta_c}(c | s^{(t)}, o^{(t)}) p(s^{(t+1)} | s^{(t)}, o^{(t)}, c), \quad (15)$$

where c denotes a shelf compartment. The compartment-level model p_{θ_c} operates on coarse geometric and physical summaries, while the local model generates precise object poses and limited rearrangements within the selected compartment.

5.1.2 Constraint-Specific Generative Modeling. Rather than learning a single monolithic generator, we represent each geometric or physical constraint by a dedicated diffusion or energy-based model. Let x denote the local configuration variables (new object pose and allowed rearrangements). We define a set of constraint experts:

$$\mathcal{E} = \{E_{\text{coll}}, E_{\text{stab}}, E_{\text{contain}}, E_{\text{move}}\},$$

where each expert corresponds to a specific constraint:

Collision-Free Expert. $E_{\text{coll}}(x)$ penalizes inter-object and object-shelf collisions. This expert may be implemented analytically using signed distance fields or as a learned energy model trained on collision-free versus colliding samples.

Stacking Stability Expert. $E_{\text{stab}}(x)$ evaluates whether stacked objects are statically stable. This expert is implemented as a learned energy-based model or classifier trained from physics simulation rollouts, assigning low energy to configurations that remain stable under perturbations.

Containment Expert. $E_{\text{contain}}(x)$ enforces that objects lie within compartment bounds and shelf openings. This expert is typically analytic, but can be learned for complex shelf geometries.

Rearrangement Effort Expert. $E_{\text{move}}(x)$ penalizes displacement of existing objects, encouraging minimal disruption.

Each expert can be trained independently using data and supervision appropriate to the constraint, enabling reuse and extension without retraining the entire system.

5.1.3 Local Generative Model and Training. We train a *base local generator* $p_{\theta_l}(x \mid s^{(t)}, o^{(t)}, c)$ that captures the distribution of plausible local placements and rearrangements without enforcing all constraints explicitly. Training data consists of locally valid configurations sampled from simulation under relaxed constraints.

The base generator can be instantiated as:

- a conditional diffusion model trained via denoising score matching, or
- a lightweight proposal network for energy-based sampling.

Crucially, this generator is trained on intermediate configurations arising from greedy insertion trajectories, ensuring alignment with inference-time states.

5.1.4 Coupling Constraint Experts at Inference. At inference time, constraint experts are composed to enforce the desired feasibility conditions. Specifically, local configurations are sampled according to:

$$p(x \mid s^{(t)}, o^{(t)}, c) \propto p_{\theta_l}(x \mid s^{(t)}, o^{(t)}, c) \prod_j \exp(-w_j E_j(x)), \quad (16)$$

where $E_j \in \mathcal{E}$ and w_j are user- or task-dependent weights.

In the diffusion setting, this corresponds to guided sampling:

$$\nabla_x \log p(x) = \nabla_x \log p_{\theta_l}(x) - \sum_j w_j \nabla_x E_j(x), \quad (17)$$

while in the EBM setting, sampling is performed via Langevin dynamics over the summed energy.

This composition allows constraints to be enabled, disabled, or reweighted at deployment time without retraining the base generator.

5.1.5 Compartment-Level Feasibility Estimation. The compartment-level generator p_{θ_c} is trained using aggregated signals from constraint experts. For each compartment, we estimate whether at least one low-energy configuration exists under bounded rearrangement. This enables p_{θ_c} to prioritize compartments that are not only semantically suitable, but also physically feasible under the active constraint set.

5.1.6 Training the Compartment-Level Generator. The compartment-level generator $p_{\theta_c}(c \mid s^{(t)}, o^{(t)})$ is trained to predict which shelf compartments can feasibly accommodate the new object under geometric and physical constraints, possibly requiring limited rearrangement. Its role is to prune the global search space before invoking local generative models.

Feasibility Label Generation. Training supervision for p_{θ_c} is obtained by querying the constraint-specific experts at the local level. For each training state $(s^{(t)}, o^{(t)})$ and each compartment $c \in C$, we estimate a compartment feasibility score:

$$F(c \mid s^{(t)}, o^{(t)}) = \min_{x \in \mathcal{X}_c} \sum_{j \in \mathcal{E}} w_j E_j(x), \quad (18)$$

where:

- $\mathcal{X}_c$ denotes the space of local configurations restricted to compartment c ,
- $E_j \in \mathcal{E}$ are the constraint-specific energy experts (collision, stability, containment, rearrangement),
- w_j are fixed or sampled constraint weights.

In practice, $F(c \mid s^{(t)}, o^{(t)})$ is approximated by sampling a small number of local configurations using the base generator p_{θ_l} and evaluating their energies.

Supervision Targets. From the estimated feasibility score, we derive training targets for each compartment:

- **Binary labels:**

$$y_c = \mathbb{I}[F(c \mid s^{(t)}, o^{(t)}) < \tau],$$

indicating whether at least one low-energy placement exists;

- **Soft targets:**

$$y_c = \exp(-F(c \mid s^{(t)}, o^{(t)})),$$

encoding relative feasibility across compartments;

- **Cost-aware targets:**

$$y_c = \exp(-F(c \mid s^{(t)}, o^{(t)}) - \beta C_{\text{move}}(c)),$$

incorporating estimated rearrangement effort.

These targets allow the compartment-level model to learn not only feasibility, but also relative preference among feasible compartments.

Training Objective. The compartment-level generator is trained using a multi-label classification or regression objective:

$$\mathcal{L}_{\text{comp}} = \sum_{c \in C} \ell(p_{\theta_c}(c \mid s^{(t)}, o^{(t)}), y_c), \quad (19)$$

where ℓ is binary cross-entropy or squared loss depending on the target formulation.

Alignment with Hierarchical Inference. Because feasibility labels are derived from the same constraint experts used at inference time, p_{θ_c} learns to anticipate the downstream behavior of the local diffusion / EBM models. This ensures that compartments ranked highly during inference are precisely those likely to admit low-energy, constraint-satisfying placements with minimal rearrangement.

Semantic Decoupling. Importantly, p_{θ_c} is trained solely on geometric and physical feasibility. Semantic preferences are applied only during greedy selection among feasible candidates, preserving modularity and enabling reuse across users.

5.1.7 Hierarchical Inference Procedure. For each greedy insertion step, hierarchical inference proceeds as follows:

- (1) Sample or rank feasible compartments using p_{θ_c} .
- (2) For each candidate compartment, sample local configurations using the base generator p_{θ_l} .
- (3) Apply constraint experts via guided diffusion or energy minimization.
- (4) Score feasible candidates using semantic distance, preference diffusion, and movement cost.
- (5) Select the best-scoring configuration as $s^{(t+1)}$.

5.1.8 Advantages of Constraint-Specific Composition. This design offers several advantages:

- **Modularity:** new constraints (e.g., fragility, accessibility) can be added as new experts.
- **Data efficiency:** each expert is trained only on relevant data.
- **Flexibility:** constraints can be reweighted or toggled at inference time.
- **Interpretability:** violations can be traced to specific constraint energies.

By combining hierarchical generation with constraint-specific diffusion and energy-based models, the proposed framework achieves scalable, flexible, and physically grounded shelf arrangement under greedy insertion.

5.2 Training as Incremental Insertion

Rather than training on complete shelf configurations in isolation, we model shelf organization as a sequence of insertion steps. Starting from an initial configuration $s^{(0)} = s_{\text{init}}$, we construct trajectories:

$$s^{(0)} \rightarrow s^{(1)} \rightarrow \dots \rightarrow s^{(T)},$$

where each transition $s^{(t)} \rightarrow s^{(t+1)}$ inserts exactly one new object and optionally rearranges existing ones.

Each trajectory corresponds to a realistic usage pattern in which objects arrive over time. This formulation eliminates train–test mismatch and prevents exposure bias when greedy insertion is used at inference.

5.3 Synthetic Data Generation

We generate training data using a physics-aware simulator and procedural shelf generator. Shelves vary in the number of layers, compartment sizes, and support surfaces. Objects are sampled from diverse shape and category distributions and assigned semantic attributes (e.g., usage frequency, fragility).

Training scenes include:

- empty-shelf initialization scenes,
- incremental insertion scenes with limited rearrangement,

- dense clutter scenes requiring non-trivial space creation.

All generated states are filtered to ensure membership in the geometrically valid subspace S_{valid} .

5.4 Candidate Generation per Greedy Step

At each step t , given the current configuration $s^{(t)}$ and the object $o^{(t)}$ to be inserted, we generate a set of candidate next configurations:

$$\{s_i^{(t+1)}\}_{i=1}^K \sim p_\theta(s \mid s^{(t)}, o^{(t)}),$$

where p_θ is a diffusion-based or energy-based generative model.

This inner-loop generator enforces hard geometric and physical constraints, including:

- collision avoidance,
- containment within shelf geometry,
- static stability,
- bounded rearrangement of existing objects.

As a result, greedy insertion never considers infeasible candidates.

5.5 Stepwise Semantic Preservation

Semantic preservation is enforced locally at every insertion step. For each candidate transition $s^{(t)} \rightarrow s^{(t+1)}$, we define a stepwise semantic loss:

$$L_{\text{sem}}^{(t)} = D_{\theta_2}(s^{(t+1)}, s^{(t)}) - \alpha g_{\theta_3}(f_{\theta_1}(s^{(t+1)})), \quad (20)$$

which penalizes semantic drift while encouraging compliance with user preferences.

By minimizing semantic deviation incrementally, the framework preserves global structure across long insertion sequences without requiring foresight over the entire task horizon.

5.6 Movement Minimization via Displacement Regularization

To discourage unnecessary rearrangement, we introduce a stepwise movement penalty:

$$L_{\text{move}}^{(t)} = \sum_{i=1}^{n_t} \|p_i^{(t+1)} - p_i^{(t)}\|, \quad (21)$$

where the sum ranges over objects already present at step t . This term encourages the generative model to create space using minimal, local adjustments rather than global reshuffling.

5.7 Joint Stepwise Training Objective

For each insertion step, we define the total training loss:

$$L^{(t)} = L_{\text{geo}}^{(t)} + \lambda_{\text{sem}} L_{\text{sem}}^{(t)} + \lambda_{\text{move}} L_{\text{move}}^{(t)}, \quad (22)$$

where $L_{\text{geo}}^{(t)}$ enforces geometric and physical validity. The model parameters are optimized to minimize the expected loss over insertion trajectories.

5.8 Preference Diffusion and Continual Learning

Semantic alignment is further shaped using preference diffusion. Given a trajectory $\tau = (s^{(0)}, \dots, s^{(T)})$, we model:

$$p_\phi(\text{pref} \mid \tau), \quad s_\phi(\tau) = \nabla_\tau \log p_\phi(\text{pref} \mid \tau),$$

which provides smooth, reward-like semantic gradients across insertion steps.

User feedback is incorporated online using a continual learning ODE with replay and regularization, enabling adaptation to evolving preferences while maintaining stability.

5.9 LLM-Assisted Semantic Supervision

Large language models are used to improve data efficiency during training. Specifically, LLMs generate:

- weak semantic labels for configurations,
- pairwise preference comparisons,
- challenging negative examples for contrastive learning.

These signals are treated as soft supervision and are refined through preference diffusion and limited human feedback.

5.10 Adversarial Semantic Regularization

To encourage global semantic realism, we introduce a semantic quality discriminator Q_ω operating on semantic embeddings. The adversarial loss

$$L_{\text{adv}} = -\log Q_\omega(f_{\theta_1}(s^{(t+1)}))$$

penalizes semantically implausible configurations and complements the stepwise semantic loss by enforcing global consistency.

5.11 Relation to Reinforcement Learning

The training process can be interpreted as a finite-horizon Markov decision process, where states are shelf configurations and actions correspond to compartment selection, rearrangement, and placement. Unlike standard reinforcement learning, we avoid explicit value estimation and unstable reward bootstrapping. Instead, we rely on dense stepwise losses, generative proposals, and preference gradients, achieving stable optimization with strong long-horizon consistency.

6 Advantages of the Proposed Method

The proposed framework introduces a hierarchical and compositional formulation for robotic shelf organization that jointly addresses geometric feasibility, physical validity, semantic alignment, and user preference adaptation. We highlight several key advantages that distinguish our approach from prior work.

6.1 Hierarchical Decomposition Reduces Combinatorial Complexity

Shelf organization is inherently hierarchical: humans first decide which shelf compartment an object belongs to, and only then determine its precise placement. By explicitly decomposing generation into a global compartment-level decision followed by local pose generation, our method dramatically reduces the effective search space. This decomposition enables efficient sampling, improves data efficiency, and stabilizes greedy insertion, particularly in densely populated shelves where flat generation would be intractable.

6.2 Composable Constraint Modeling Enables Modularity and Flexibility

Rather than relying on a single monolithic generator, we model each geometric or physical constraint—such as collision avoidance, stacking stability, containment, and rearrangement effort—with a dedicated diffusion or energy-based expert. These constraint-specific models can be trained independently using appropriate supervision (analytic geometry, physics simulation, or weak labels), and composed only at inference time. This modular design allows constraints to be added, removed, or reweighted without retraining the entire system, supporting flexible deployment across different shelf designs, task requirements, and user preferences.

6.3 Separation of Feasibility and Preference Improves Generalization

The framework explicitly decouples *feasibility* from *preference*. Geometric and physical constraints are enforced by generative and energy-based models that define a feasible configuration manifold, while semantic alignment and user habits are handled by a separate semantic distance and preference diffusion mechanism. This separation prevents overfitting semantics to specific geometries and enables the system to generalize across novel shelf layouts, object sets, and users with minimal additional data.

6.4 Greedy Insertion with Stepwise Regularization Preserves Global Semantics

Although the system inserts objects greedily, semantic coherence is preserved through stepwise regularization. At each insertion step, semantic distance to the current configuration and explicit movement penalties are minimized, preventing large semantic jumps or unnecessary rearrangements. Because both the generative models and semantic embedding are trained on intermediate greedy states, greedy inference does not induce distribution shift and yields globally consistent configurations over long insertion sequences.

6.5 Data Efficiency through Locality and Weak Supervision

The hierarchical and compositional structure significantly improves data efficiency. Local generators focus on compact subspaces defined by individual compartments, while constraint experts are trained only on data relevant to their respective physical properties. Semantic alignment relies on relative judgments and weak supervision, which can be obtained from large language models or limited human feedback. As a result, the system avoids the need for large-scale fully annotated datasets while maintaining strong performance.

6.6 Preference Diffusion Enables Continual Personalization

User preferences are modeled using preference diffusion, which provides reward-like gradient guidance without requiring explicit scalar reward engineering. This formulation naturally supports continual learning: preferences can be updated online as new feedback arrives, while replay and regularization prevent catastrophic forgetting. The resulting system adapts over time to individual users while preserving stable behavior.

6.7 Interpretability and Debuggability

Each component of the framework has a clear semantic role. Failures can be localized to the compartment selector, a specific constraint expert, or the semantic alignment module. Constraint energies provide interpretable signals indicating why a candidate placement was rejected, facilitating debugging, user feedback, and trust in human-robot interaction.

6.8 Scalability and Practical Deployment

Finally, the proposed method scales naturally to shelves with many compartments and objects. Inference consists of hierarchical sampling with lightweight constraint composition and semantic scoring, making it suitable for real-time or interactive systems. Because the framework produces goal configurations rather than low-level control commands, it integrates seamlessly with existing motion planning and execution pipelines.

Together, these advantages make the proposed approach a robust, flexible, and data-efficient solution for personalized robotic shelf organization under geometric, physical, and semantic constraints.

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